Complex Analysis Homework 1

1) Let $z = \cos\theta + i\sin\theta$.

(i) Show that $z^n + z^{-n} = 2\cos n\theta$, where n is a positive integer.

(ii) Let m be a positive integer. Show that

$$(2\cos\theta)^{2m} = 2\left[\cos 2m\theta + \binom{2m}{1}\cos(2m-2)\theta + \binom{2m}{2}\cos(2m-4)\theta + \cdots + \binom{2m}{m-1}\cos 2\theta\right] + \binom{2m}{m}.$$

(iii) Hence, or otherwise, prove that

$$\int_0^{\frac{\pi}{2}} \cos^{2m}\theta\, d\theta = \frac{\pi}{2^{2m+1}}\binom{2m}{m}$$

where m is a positive integer.

Solution

(i) $z = \cos\theta + i\sin\theta$

$z^n = \cos n\theta + i\sin n\theta$

$z^{-n} = \cos n\theta - i\sin n\theta$

$z^n + z^{-n} = 2\cos n\theta$

(ii) $(2\cos\theta)^{2m} = \left(z + \frac{1}{z}\right)^{2m}$

$$z^{2m} + \binom{2m}{1} z^{2m-2} + \ldots + \binom{2m}{m} + \ldots + \binom{2m}{2m-1} z^{-2m+2} + z^{-2m}$$

$$= \left(z^{2m} + z^{-2m}\right) + \binom{2m}{1}\left(z^{2m-2} + z^{-2m+2}\right) + \ldots + \binom{2m}{m}$$

$$= 2\cos 2m\theta + \binom{2m}{1}.2\cos(2m-2)\theta + \ldots + \binom{2m}{m-1} 2\cos 2\theta + \binom{2m}{m}$$

as required.

(iii) $\int_0^{\pi/2} \cos 2k\theta d\theta = \left[\frac{\sin 2k\theta}{2k}\right]_0^{\pi/2} = 0$ if $k > 0$

$$\int_0^{\pi/2} (2\cos\theta)^{2m} d\theta = \binom{2m}{m} \int_0^{\pi/2} 1 d\theta = \frac{\pi}{2}\binom{2m}{m}$$

$$\int_0^{\pi/2} \cos^{2m}\theta d\theta = \frac{\pi}{2^{2m+1}}\binom{2m}{m}$$

2. Let n be a positive integer.

(a) Assume that θ is not an integral multiple of π.

(i) Prove that

$$\frac{(\cos\theta + i\sin\theta)^n - 1}{\cos\theta + i\sin\theta - 1} = \frac{\sin\frac{n\theta}{2}}{\sin\frac{\theta}{2}}\left(\cos\frac{(n-1)\theta}{2} + i\sin\frac{(n-1)\theta}{2}\right).$$

(ii) Using the identity $\sum_{k=1}^{n} z^k = z\left(\frac{z^n - 1}{z - 1}\right)$ for $z \neq 1$, or otherwise, prove that $\sum_{k=1}^{n} \cos k\theta = \frac{\sin\frac{n\theta}{2}\cos\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$ and

$$\sum_{k=1}^{n} \sin k\theta = \frac{\sin\frac{n\theta}{2}\sin\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}.$$

(iii) Using (a)(ii), or otherwise, prove that

$$\sum_{k=1}^{n} \cos^2(k\theta) = \frac{n}{2} + \frac{\sin n\theta\cos(n+1)\theta}{2\sin\theta}.$$

(b) For $n > 1$, evaluate

(i) $\sum_{k=1}^{n} \sin^2\left(\frac{k\pi}{n}\right)$,

(ii) $\sum_{k=1}^{n} \left(\sin\frac{k\pi}{n} + \cos\frac{k\pi}{n}\right)^2$.

Solution

2. (a) (i)
$$\frac{(\cos\theta+i\sin\theta)^n-1}{\cos\theta+i\sin\theta-1}$$
$$=\frac{\cos n\theta-1+i\sin n\theta}{\cos\theta-1+i\sin\theta}$$
$$=\frac{-2\sin^2\frac{n\theta}{2}+2i\sin\frac{n\theta}{2}\cos\frac{n\theta}{2}}{-2\sin^2\frac{\theta}{2}+2i\sin\frac{\theta}{2}\cos\frac{\theta}{2}}$$
$$=\frac{2i\sin\frac{n\theta}{2}\left(\cos\frac{n\theta}{2}+i\sin\frac{n\theta}{2}\right)}{2i\sin\frac{\theta}{2}\left(\cos\frac{\theta}{2}+i\sin\frac{\theta}{2}\right)}$$
$$=\frac{\sin\frac{n\theta}{2}}{\sin\frac{\theta}{2}}\left(\cos\frac{(n-1)\theta}{2}+i\sin\frac{(n-1)\theta}{2}\right)$$

(ii) Putting $z=\cos\theta+i\sin\theta$ in $\sum_{k=1}^{n}z^k=z\left(\frac{z^n-1}{z-1}\right)$, we have

$$\sum_{k=1}^{n}(\cos\theta+i\sin\theta)^k=(\cos\theta+i\sin\theta)\left(\frac{(\cos\theta+i\sin\theta)^n-1}{\cos\theta+i\sin\theta-1}\right).$$

So, by (a)(i), we have

$$\sum_{k=1}^{n}(\cos\theta+i\sin\theta)^k=(\cos\theta+i\sin\theta)\left(\frac{\sin\frac{n\theta}{2}}{\sin\frac{\theta}{2}}\right)\left(\cos\frac{(n-1)\theta}{2}+i\sin\frac{(n-1)\theta}{2}\right)$$

$$\sum_{k=1}^{n}(\cos k\theta+i\sin k\theta)=\frac{\sin\frac{n\theta}{2}\cos\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}+i\frac{\sin\frac{n\theta}{2}\sin\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$$

$$\sum_{k=1}^{n}\cos k\theta+i\sum_{k=1}^{n}\sin k\theta=\frac{\sin\frac{n\theta}{2}\cos\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}+i\frac{\sin\frac{n\theta}{2}\sin\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$$

By comparing the real parts and the imaginary parts of the both sides, we have

$$\sum_{k=1}^{n}\cos k\theta=\frac{\sin\frac{n\theta}{2}\cos\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}\quad\text{and}\quad\sum_{k=1}^{n}\sin k\theta=\frac{\sin\frac{n\theta}{2}\sin\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}.$$

(iii) $\displaystyle\sum_{k=1}^{n}\cos^2(k\theta)$

$$=\sum_{k=1}^{n}\frac{1+\cos 2k\theta}{2}$$

$$=\frac{n}{2}+\frac{1}{2}\sum_{k=1}^{n}\cos 2k\theta$$

$$=\frac{n}{2}+\frac{\sin n\theta\cos(n+1)\theta}{2\sin\theta}$$ (by (a)(ii))

(b) (i) $\sum_{k=1}^{n} \sin^2\left(\frac{k\pi}{n}\right)$

$$= \sum_{k=1}^{n} \left(1 - \cos^2\left(\frac{k\pi}{n}\right)\right)$$

$$= n - \sum_{k=1}^{n} \cos^2\left(\frac{k\pi}{n}\right)$$

$$= n - \left(\frac{n}{2} + \frac{\sin\pi\cos\frac{(n+1)\pi}{n}}{2\sin\frac{\pi}{n}}\right) \quad \text{(by (a)(iii))}$$

$$= n - \frac{n}{2}$$

$$= \frac{n}{2}$$

$$\sum_{k=1}^{n} \sin^2\left(\frac{k\pi}{n}\right)$$

$$= \sum_{k=1}^{n} \frac{1 - \cos\frac{2k\pi}{n}}{2}$$

$$= \frac{n}{2} - \frac{1}{2}\sum_{k=1}^{n} \cos\frac{2k\pi}{n}$$

$$= \frac{n}{2} - \frac{\sin\pi\cos\frac{(n+1)\pi}{n}}{2\sin\frac{\pi}{n}} \quad \text{(by (a)(ii))}$$

$$= \frac{n}{2} - 0$$

$$= \frac{n}{2}$$

(ii) $$\sum_{k=1}^{n}\left(\sin\frac{k\pi}{n}+\cos\frac{k\pi}{n}\right)^2$$

$$=\sum_{k=1}^{n}\left(1+2\sin\frac{k\pi}{n}\cos\frac{k\pi}{n}\right)$$

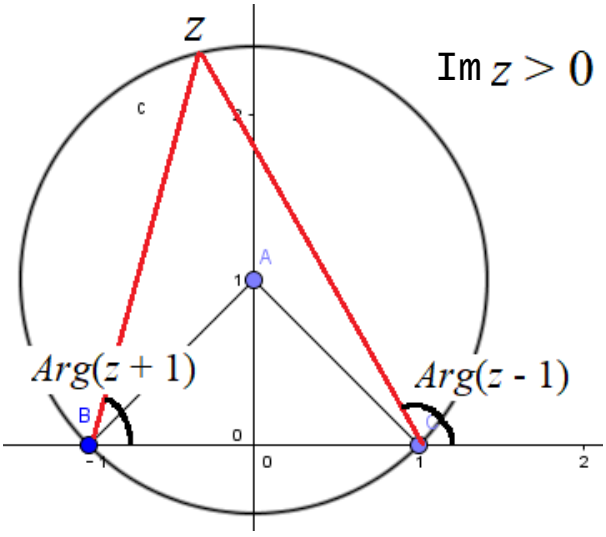
$$=n+\sum_{k=1}^{n}\sin\frac{2k\pi}{n}$$

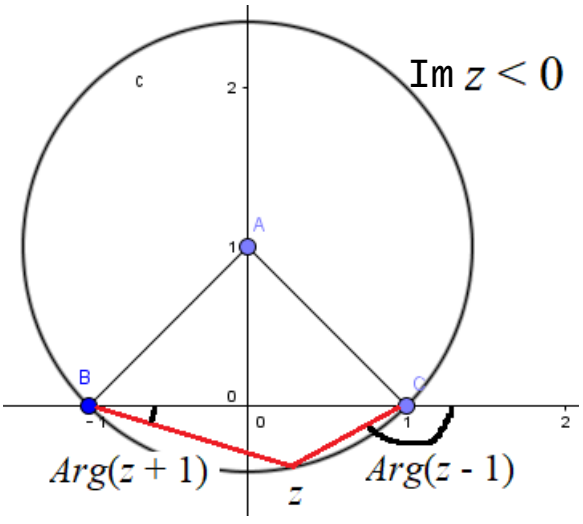
$$=n+\frac{\sin\pi\,\sin\frac{(n+1)\pi}{n}}{\sin\frac{\pi}{n}}\quad\text{(by (a)(ii))}$$

$$=n$$

3. Let z be a complex number such that $|z-i|=\sqrt{2}$ and $z\neq\pm1$. Calculate $Arg\left(\frac{z+1}{z-1}\right)$.

Solution





$$Arg(z+1)-Arg(z-1)=\begin{cases}-\frac{\pi}{4} & \text{if } \operatorname{Im} z>0\\ \frac{3\pi}{4} & \text{if } \operatorname{Im} z<0\end{cases}$$

$$Arg\left(\frac{z+1}{z-1}\right) = Arg(z+1) - Arg(z-1) + 2n\pi$$

$$-\pi < Arg\left(\frac{z+1}{z-1}\right) \leq \pi \quad \Rightarrow \quad n = 0$$

$$Arg\left(\frac{z+1}{z-1}\right) = Arg(z+1) - Arg(z-1) = \begin{cases} -\frac{\pi}{4} & \text{if } \operatorname{Im} z > 0 \\ \frac{3\pi}{4} & \text{if } \operatorname{Im} z < 0 \end{cases}$$

4. Let m be a positive integer.

(a) By using De Moivre's theorem, show that

$$\sin(2m+1)\theta = \binom{2m+1}{1}\cos^{2m}\theta\sin\theta - \binom{2m+1}{3}\cos^{2m-2}\theta\sin^3\theta + \cdots + (-1)^m\sin^{2m+1}\theta.$$

(b) Deduce that the polynomial $p(x) = \binom{2m+1}{1}x^m - \binom{2m+1}{3}x^{m-1} + \cdots + (-1)^m$ has m distinct roots $\alpha_k = \cot^2\left(\frac{k\pi}{2m+1}\right)$ where $k = 1, 2, \ldots, m$.

(c) Prove that $\cot^2\left(\frac{\pi}{2m+1}\right) + \cot^2\left(\frac{2\pi}{2m+1}\right) + \cdots + \cot^2\left(\frac{m\pi}{2m+1}\right) = \frac{m(2m-1)}{3}$.

(d) Show that $\sin\theta < \theta < \tan\theta$ for $0 < \theta < \frac{\pi}{2}$. Deduce that

$$\lim_{m\to\infty}\left(\frac{1}{1^2} + \frac{1}{2^2} + \cdots + \frac{1}{m^2}\right) = \frac{\pi^2}{6}$$

Solution

(a) By using De Moivre's theorem, we have

$$\cos(2m+1)\theta + i\sin(2m+1)\theta = (\cos\theta + i\sin\theta)^{2m+1}$$

Consider even power of $\cos\theta$ multiply by odd power of $i\sin\theta$,

$$i\sin(2m+1)\theta=\binom{2m+1}{1}\cos^{2m}\theta(i\sin\theta)+\binom{2m+1}{3}\cos^{2m-2}\theta(i\sin\theta)^3+\cdots+(i\sin\theta)^{2m+1}$$

$$=i\binom{2m+1}{1}\cos^{2m}\theta\sin\theta-i\binom{2m+1}{3}\cos^{2m-2}\theta\sin^3\theta+\cdots+(-1)^m i\sin^{2m+1}\theta.$$

$$\sin(2m+1)\theta=\binom{2m+1}{1}\cos^{2m}\theta\sin\theta-\binom{2m+1}{3}\cos^{2m-2}\theta\sin^3\theta+\cdots+(-1)^m\sin^{2m+1}\theta.$$

(b) $$\frac{\sin(2m+1)\theta}{\sin^{2m+1}\theta}=\binom{2m+1}{1}\cot^{2m}\theta-\binom{2m+1}{3}\cot^{2m-2}\theta+\cdots+(-1)^m=p(\cot^2\theta)\ \text{ if }\ \sin\theta\neq 0.$$

$\theta=\dfrac{k\pi}{2m+1}$, k = 1, 2,…, m, are distinct zeros of $\sin(2m+1)\theta$ with $\sin\theta\neq 0$. Therefore $\alpha_k=\cot^2\left(\dfrac{k\pi}{2m+1}\right)$ where k = 1, 2,…,m, m distinct roots of $p(x)$.

(c) $$p(x)=\binom{2m+1}{1}x^m-\binom{2m+1}{3}x^{m-1}+\cdots+(-1)^m=\binom{2m+1}{1}(x-\alpha_1)(x-\alpha_2)\cdots(x-\alpha_m)$$

Comparing the coefficient of x^{m-1}, we have

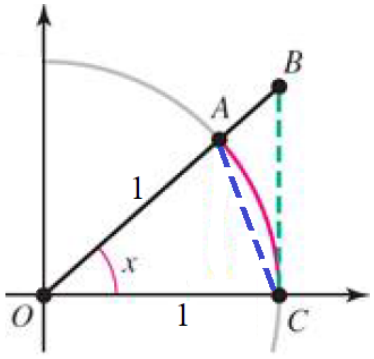
$$-\binom{2m+1}{3}=-\binom{2m+1}{1}(\alpha_1+\alpha_2+\cdots+\alpha_m)$$

$$\cot^2\left(\frac{\pi}{2m+1}\right)+\cot^2\left(\frac{2\pi}{2m+1}\right)+\cdots+\cot^2\left(\frac{m\pi}{2m+1}\right)=\binom{2m+1}{3}\Big/\binom{2m+1}{1}=\frac{m(2m-1)}{3}.$$

(d) Area of $\Delta OAC=\dfrac{1}{2}\cdot 1^2\cdot\sin x$

Area of sector $OAC=\dfrac{1}{2}\cdot 1^2\cdot x=\dfrac{x}{2}$

Area of $\Delta OBC=\dfrac{1}{2}(OC)(BC)=\dfrac{1}{2}\tan x$



Since area of $\Delta OAC<$ area of sector $OAC<$ area of ΔOBC, we have

$$\sin\theta<\theta<\tan\theta$$

Hence, $\cot\theta < \dfrac{1}{\theta} < \csc\theta$

$$\cot^2\left(\frac{\pi}{2m+1}\right)+\cdots+\cot^2\left(\frac{m\pi}{2m+1}\right)<\left(\frac{2m+1}{\pi}\right)^2+\cdots+\left(\frac{2m+1}{m\pi}\right)^2<\csc^2\left(\frac{\pi}{2m+1}\right)+\cdots+\csc^2\left(\frac{m\pi}{2m+1}\right)$$

$$\frac{m(2m-1)}{3}<\left(\frac{2m+1}{\pi}\right)^2\left(\frac{1}{1^2}+\frac{1}{2^2}+\cdots+\frac{1}{m^2}\right)<\frac{m(2m-1)}{3}+m=\frac{m(2m+2)}{3}$$

$$\frac{m(2m-1)}{3}\left(\frac{\pi}{2m+1}\right)^2<\frac{1}{1^2}+\frac{1}{2^2}+\cdots+\frac{1}{m^2}<\frac{m(2m+2)}{3}\left(\frac{\pi}{2m+1}\right)^2$$

$$\lim_{m\to\infty}\left(\frac{1}{1^2}+\frac{1}{2^2}+\cdots+\frac{1}{m^2}\right)=\frac{\pi^2}{6}$$

5. Suppose that $z=\dfrac{1}{2}\left(\cos\theta+i\sin\theta\right)$.

(a) Show that $1+z+z^2+\cdots=\dfrac{1}{1-z}$.

(b) Express $1+\dfrac{1}{2}\cos\theta+\dfrac{1}{2^2}\cos 2\theta+\dfrac{1}{2^3}\cos 3\theta+\cdots$ in terms of $\cos\theta$.

(c) Calculate $\arg\left(1+z+z^2+\cdots\right)$ for $\theta=-\dfrac{\pi}{3}$.

Solution

(a) $\lim\limits_{n\to\infty}\left|z^n\right|=\lim\limits_{n\to\infty}\dfrac{1}{2^n}=0 \quad\Rightarrow\quad \lim\limits_{n\to\infty}z^n=0$

$$1+z+z^2+\cdots=\lim_{n\to\infty}\left(1+z+z^2+\cdots+z^{n-1}\right)=\lim_{n\to\infty}\frac{1-z^n}{1-z}=\frac{1}{1-z}.$$

(b) $$\frac{1}{1-z}=\frac{2}{2-\cos\theta-i\sin\theta}=\frac{2-\cos\theta+i\sin\theta}{(2-\cos\theta)^2+\sin^2\theta}=2\frac{2-\cos\theta}{5-4\cos\theta}+2i\frac{\sin\theta}{5-4\cos\theta}$$

$$1+\frac{1}{2}\cos\theta+\frac{1}{2^2}\cos 2\theta+\frac{1}{2^3}\cos 3\theta+\cdots=\mathrm{Re}\left(1+z+z^2+\cdots\right)=\mathrm{Re}\left(\frac{1}{1-z}\right)=2\frac{2-\cos\theta}{5-4\cos\theta}.$$

(c) $\arg\left(\dfrac{1}{1-z}\right)=\arg\left(1-\dfrac{1}{\sqrt{3}}i\right)=-\dfrac{\pi}{6}+2k\pi,$ where k is any integer.